PEEU: Explore Websites, Relabel Tasks with Hindsight, Train Small Agents to Plan

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Paper version: arXiv 2606.27330v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *they suffer from weak planning and limited cross-website generalization*

Small open-source MLLMs are cheap and private but plan weakly. Atomic tasks don't compose upward; coarse tasks drift from what the agent actually did.

Why hindsight

1. **Atomic tasks.** Do click/type skills compose? Unclear.
2. **Coarse tasks.** Task says 4.5 stars, run reaches 4.
3. **Hindsight.** Relabel the task to what the run did.

Evaluation setup

WebVoyager · online · 7 held-out sites · 0.1k / 2k trajectories · Qwen2.5-VL-3B/7B · GPT-4o explorer · 15-step budget · 4×A800

- **Online** success rate; TDHAF offline, kept apart.
- **Held out:** 7 test sites never explored or trained.

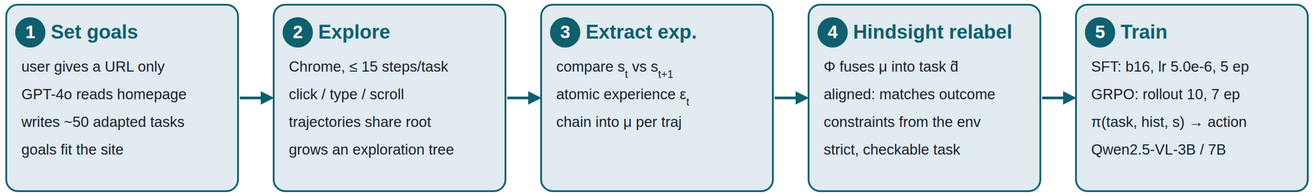
Our read (CUA Radar)

61	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval (WebVoyager SR); 7 held-out sites; strong baselines at the same data scale; granularity ablation (TDHAF); 4×A800.
- **Not reported:** seeds/std, judge validation, code. **Flags:** no login/CAPTCHA/payment; AP8 not flagged.
- Pattern K14. Score and band from CUA Radar; authors' numbers, not rerun.

Method: PEEU — explore, then relabel the task to the trajectory

- **Sees / acts:** SoM-labeled screenshot + page text in; click, type, scroll, wait, go back, answer out — Chrome, live sites.
- **Hindsight relabel:** GPT-4o reads before/after states, writes atomic experiences ε_t, fuses them into a task the trajectory actually satisfies, with constraints only exploration can find.
- **Signal:** SFT on (relabeled task, trajectory) pairs; GRPO adds format + answer rewards (2.0 max).



Our redraw (not the paper's figure): set goals, explore, extract experiences, hindsight-relabel, train.

$$\tilde{D} = \Phi(\mu_1, \ldots, \mu_n; M)$$

aggregate trajectory experiences μ into hindsight-aligned PEEU tasks

Key results: WebVoyager online success (%)

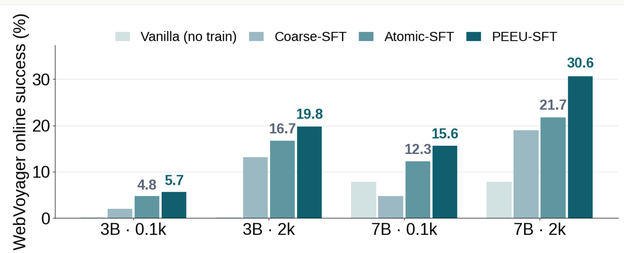


Table 1, redrawn: overall = average over ID site + 7 unseen sites; one run per cell, no seeds.

PEEU-SFT 7B: 30.6% vs 22.7% for Qwen2.5-VL-32B Instruct

SO WHAT → *GPT-4o vanilla reaches 59.0 on the same benchmark.*

TDHAF: task granularity vs generalization

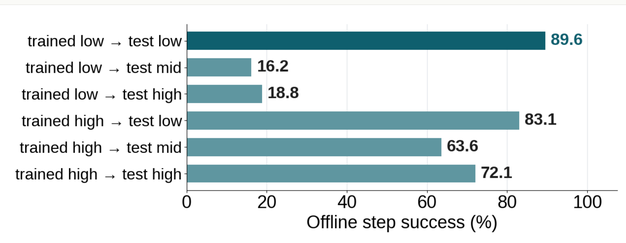


Table 2, redrawn: offline Step SR on Multimodal-Mind2Web-derived data (7B, in-domain).

SO WHAT → *Low-level mastery does not reach high-level planning (89.6 → 18.8); high-level training covers every level. OOD coverage, 7B: 37.8 (high-trained) vs 25.7 (low-trained); 3B: 33.8 vs 18.9. Coverage counts samples correct at all three levels.*

Headline Numbers

30.6%
WebVoyager overall
7B · 2k trajectories

22.7%	Qwen2.5-VL-32B
21.7%	Atomic-SFT, 2k
19.0%	Coarse-SFT, 2k

The paper's own controls

CONTROL	VALUE	CONTEXT
Data scale	0.1k / 2k traj	same for all methods
Held-out sites	7 sites unseen	train: Allrecipes + 2k
GRPO, 7B, 0.1k	PEEU 19.9 / Atomic 10.0	Coarse 11.5
Offline (TDHAF)	Step SR offline	616 / 684 samples

Takeaway

Hindsight high-level tasks beat coarse and atomic labels at equal data; low-level skill alone does not give planning.

GPT-4o · 15-step · Qwen2.5-VL-3B / 7B · SFT + GRPO · 7 held-out sites · no seeds

Scan to read the full paper

arxiv.org/abs/2606.27330v1